

One Timestamp Does Not Identify Temporal Support in Event-Detection Benchmarks

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Abstract

001 Event-detection benchmarks compare a prediction with
002 a label at a single nominal timestamp, and that times-
003 tamp does not identify when the evidence supporting
004 the prediction was acquired or which instant its out-
005 put represents. This paper measures two predictor-
006 side quantities and establishes a complementary limit
007 on attributing within-exposure event timing to object
008 timing using released artifacts. For a released recur-
009 rent detector, the newest 50 ms input window has an
010 occlusion-sensitivity centroid 23.81 ms before the label
011 instant, while the regression-based temporal coefficient
012 estimated on nominal-time-matched detections with
013 centered label velocity is -2.40 ± 7.18 ms, statistically
014 consistent with zero under the reported specification.
015 Across the evaluated specifications with geometric con-
016 trols, the coefficient ranges from -20.2 to $+11.1$ ms,
017 whereas the newest-window occlusion-sensitivity cen-
018 teroid corresponds to $+23.81$ ms under the convention
019 of Eq. 2. On the velocity-evaluable subset, average pre-
020 cision is nearly invariant between the label instant and
021 the newest-window occlusion-sensitivity centroid: ap-
022 plying the corresponding center displacement changes
023 mAP by only 0.0004 absolute mAP (0.04 percentage
024 points, 0.12 % relative to 0.3346) and reorders none of
025 the 5 checkpoints at the measured centroid displace-
026 ment. On the label side, DSEC’s ceiling-exposure se-
027 quences carry a harmonic signature at 100.00 Hz, show-
028 ing that within-exposure event timing cannot be at-
029 tributed to object timing without separating illumina-
030 tion structure.

031 1. Introduction

032 An object detection benchmark evaluates a prediction
033 $\hat{s}$ against a ground truth s^* through a function $\rho(\hat{s}, s^*)$
034 of position, shape and class, indexed by a query time
035 t . The query time selects which label the prediction is

compared against. Beyond selecting the label at query
time t , the scoring function contains no variable repre-
senting temporal support or effective state time, so a
temporal discrepancy is scored as displacement. Nei-
ther side’s temporal support is specified by that nom-
inal instant. A predictor computes $\hat{s}$ from a window
of sensor data. A label is produced by an exposure,
an annotator or tracker working on the exposed image,
and a geometric transfer into another sensor’s frame.

The released input-window convention is fixed by
the detector configuration and is measurable from re-
leased artifacts. The detector’s full recurrent tempo-
ral support is not identified by that convention. The
event branch of the detector examined here [23] con-
sumes a window of 50 ms divided into ten bins, and
the preprocessing script of its own repository builds
the window boundaries by counting backwards from
label timestamps, so each window ends at the label it
is scored against (Sec. 3.2). Under uniform weighting
the evidence in such a window is centered 25 ms before
the instant the label asserts. Whether the trained net-
work’s bin sensitivities are uniform is a separate ques-
tion, and occluding each bin in turn on real validation
samples answers it: the occlusion-sensitivity centroid is
 -23.81 ms (Sec. 3.3). This defines a measured center-
displacement interval between the newest-window sen-
sitivity centroid and the label instant. It does not iden-
tify the centroid of the detector’s full recurrent support.

The training target is indexed at the label instant,
and the measurement of Sec. 3.4 is consistent with it,
at -2.40 ± 7.18 ms, which is a failure to reject and not
a demonstration that they coincide. The objective does
not require the regression-based coefficient to equal the
newest-window occlusion-sensitivity centroid. Under
the reported specification, their point estimates differ
by 26.2 ms. The velocity that measurement regresses
on is finite differenced from the same labels the resid-
ual is measured against, and a forward difference shares
their noise, returning an offset from a detector that has
none. The estimator is therefore calibrated on syn-

thetic tracks before it is used, and carries a placebo term on which no lag can load.

The second measurement concerns the same class of undeclared timing property on the label side. DSEC publishes each frame's own exposure interval, and across the eighteen training sequences whose exposure files resolve, the width varies by a factor of 127 against a frame period fixed at 50 ms, with 32.5 % of frames pinned at an auto-exposure ceiling of 14 996 μ s in six sequences (Fig. 3). Those six carry the harmonic series of a 100.00 Hz mains flicker, and between 95.4 % and 100.0 % of labeled boxes carry more power at the fundamental than at an off-frequency arm on the same events (Sec. 4). A 10 000 μ s illumination period within a 14 996 μ s exposure induces temporal structure in the event stream without object motion, and Sec. 4.4 reports what that leaves of a per-object statistic.

The single timestamp used by the metric does not specify the temporal support of either side of the comparison. We measure two predictor-side quantities and establish a complementary limit on attributing within-exposure event timing to object timing. The first is the occlusion-sensitivity profile of a released recurrent event detector. Its newest window has an occlusion-sensitivity centroid 23.81 ms before the label instant and accounts for 22.7 % of the summed window-ablation sensitivity measured over a 1000 ms horizon. The second is the regression-based temporal coefficient of the same checkpoint, estimated on nominal-time-matched detections under Eq. 2. It is estimated with a vector regression, a placebo term, and a sign-flip control on the velocity estimator. The estimate is consistent with the label instant and is 26.2 ms from the newest-window centroid. DSEC provides a complementary exposure-side attribution case study: a harmonic series at 100.00 Hz in six ceiling-exposure sequences shows that a within-box temporal statistic measured inside one published exposure window cannot, by itself, be attributed to object timing (Secs. 3.2–4).

The predictor- and label-side analyses are complementary case studies on separate released benchmarks. They do not jointly decompose one detector–label pair.

The primary claim concerns identifiability under the metric. Displacing the ground truth across the measured interval moves mAP by 0.04 points and reorders none of the 5 released checkpoints. A common displacement sweep also reveals a localized inversion between rvt-b and S5-S in the 25–50 px/s stratum. The same interval is 0.61 px of displacement at the ninetieth percentile of label speed and 1.29 px at the ninety-ninth, Tab. 2 locates the insensitivity in the speed composition of the benchmark rather than in an absence of

displacement.

We measure input-side temporal sensitivity in released event detectors. We further estimate a regression-based temporal coefficient under the reported specification and compare it with the newest-window sensitivity centroid. On the velocity-evaluable subset, standard average precision changes little over the label-to-centroid displacement. A separate DSEC case study limits attribution of within-exposure event timing to object timing.

2. Related work

Recovering a time offset from image-plane displacement. Qin and Shen [5] model a global camera–IMU time offset by shifting feature observations with image-plane velocity and jointly optimizing the offset with visual–inertial states and feature locations. Here the offset lies between a predictor's clock and a benchmark's label clock. Streaming perception [6] folds wall-clock compute latency into average precision and re-anchors any detector from box coordinates alone, later work adapts to the measured runtime delay [34], a continuous-stream framework scores the latest available output against high-rate ground truth and reports reordered rankings [29], and temporal tolerance has been argued for time-series event detection [35]. Each of those latencies is the time from an event to an output and falls as the device gets faster. Sensor latency, the illumination-dependent delay between a change and the timestamp the sensor assigns it, is measured at microseconds [28], and nighttime event imaging corrects an illumination-dependent trailing by learning a per-event timestamp calibration [24], both target the sensor, while here the released exposure metadata and event stream bound what a within-box temporal statistic computed inside one published exposure can be attributed to. Inside the citation graph of LET-3D-AP [1], a latency-aware average precision scores 3D detection at the instant a prediction becomes available on a named device [36]. That work addresses compute latency, whereas the interval measured in Sec. 3.5 is fixed by a data representation. The event–RGB frequency mismatch is named in [37], which scores with unmodified average precision.

Mao et al. [7] showed that video-detection AP can remain stable despite temporal detection delay and introduced Average Delay to expose that effect. This work examines which acquisition and label times are represented by a nominal benchmark timestamp. We measure predictor-side sensitivity and constrain within-exposure event-timing attribution using released detector and exposure artifacts.

Decomposing localization error along a privileged axis. LET-3D-AP [1] decomposes camera-only 3D localization error along the line of sight and defines an average precision tolerant of the longitudinal part, and reports a camera-to-label synchronization gap as a per-dataset interval that depends on horizontal object position. Sec. 3.4 keeps the idea of a privileged axis but replaces the sensor’s line of sight with the object’s own velocity vector, and estimates a time rather than defining a tolerant score. TIDE [8, 9] and DETAD [10] bucket detection error atemporally, so a temporal offset falls inside their localization bucket, and HOTA [11] takes the label clock as exact.

Per-object time in labels. Waymo’s camera_synced_box solves t_{capture} per object and stores the resulting box shift rather than the time [2, 3]. Miro et al. [4] measure within-frame per-object annotation offsets on Argoverse2 and TruckScenes and define a dispersion of the spatial error. Each computes a per-object time in order to remove it. The event-vision label formats carry the field: the Prophesee dtype used by Gen1, the 1 Mpx dataset and DSEC-Det has a per-box t . In the releases examined here, that field’s per-object degree of freedom is not specified. Across the 60 released DSEC-Det sequences, 390 118 boxes take 70 379 distinct values of t , 5.54 boxes per value, and those values are the 50 ms frame clock.

Event benchmarks and their exposures. Gen1 [16] labels are drawn on gray-level images assembled from asynchronous per-pixel measurements. On the 1 Mpx dataset [17] the event count inside a labeled box at its own timestamp is often too small to support detection [18]. DSEC-Det [13] labels by tracking [14] on rectified frames and warps into the event view, and Ev-3DOD [19] interpolates lower-rate annotations into high-rate 3D labels. That a frame’s exposure is long relative to event timing has been stated: Event-based deblurring [20] argues that accumulating events from a single frame timestamp loses information, deblurring work names auto-exposure as why the width is unknown [21], and RENet [22] uses a DSEC exposure interval as its event window while describing it as short. The cited works do not read the released per-frame widths or measure what is inside one. State space backbones [25] substantially reduce sensitivity to inference-time window duration, and adaptive spatial-temporal windows [27] retrain a detector under each of nine partitioning strategies and report the score of each. Both treat the window as a design variable to choose. Here, the released window configuration is treated as a measured property of the released detector and evaluation

pipeline. That the released window is causal is already in print: LEOD [26] states that event detectors take only events triggered before the query time, and builds a time-flip augmentation on the consequence that forward inference favors receding objects. It does not measure where inside that window the evidence sits. Protocol work supplies the standards these measurements are held to [30–33]: a repaired protocol can reverse a field’s conclusion, and a reported score is an estimate with a variance. A protocol audit can expose failures in what a benchmark identifies even when the resulting correction reorders no existing method. Average precision is nearly invariant over the label-to-centroid interval, while the regression places the coefficient away from the occlusion-sensitivity centroid under the reported specification.

3. Comparison of input sensitivity and regression-based output timing

3.1. Definition

Let a predictor P emit a state at query time t . A signed effective-weight model uses bin centers c_k and weights a_k with $\sum_k a_k = 1$, giving the effective timestamp offset $\tau_P = \sum_k a_k c_k$ under constant velocity. The empirical ablation profile instead assigns non-negative sensitivities $s_k \geq 0$ to bins and has centroid $c_s = \sum_k s_k c_k / \sum_k s_k$. These are distinct quantities: zero-ablation sensitivity depends on both the predictor and the input, and is not in general a decomposition of temporal weights. Sections 3.2–3.3 measure this operational input-sensitivity profile over the newest window. Section 3.4 estimates the regression coefficient τ under Eq. 2. Interpreting $-\tau$ as an effective state-time offset requires the absence of residual velocity-proportional spatial bias. Section 3.5 compares $-c_s$ with the regression coefficient τ , equivalently c_s with the effective timestamp $\tau_P = -\tau$.

3.2. The window and the instant it is indexed by

Two released files fix the input side. The dataset configuration config/dataset/gen1.yaml of the detector’s own repository [23] names the representation stacked_histogram_dt=50_nbins=10, a window of 50 ms divided into ten bins of 5 ms. Line 405 of scripts/genx/preprocess_dataset.py constructs the array ev_repr_timestamps_us_end by counting backwards from a label timestamp in steps of the window duration, and extends it between consecutive label timestamps. Each representation is therefore indexed by the end of its window, and those ends are placed on label times. A window attached to a label at time t holds the events of $[t - 50 \text{ ms}, t]$, whose uniform weight-

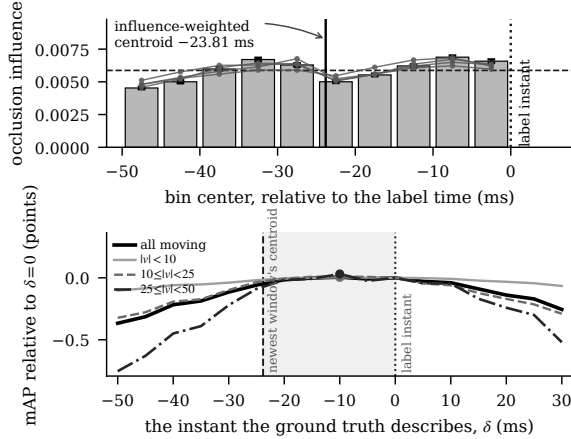


Figure 1. The newest-window sensitivity centroid differs in point estimate from the regression-based temporal coefficient, while average precision on the velocity-evaluable subset is nearly invariant between the label instant and that centroid. (a) Occlusion influence of each of the ten bins of the released rvt-t input window, over 480 real Gen1 validation samples with the recurrent state warmed eight steps, with descriptive sample-level standard errors. The dashed line is the mean over bins, the dotted line the label instant, and the solid line the occlusion-sensitivity centroid. Gray lines show the four other checkpoints, each rescaled to its own mean. (b) Average precision against the temporal center displacement δ applied to the ground truth boxes, by object speed in px/s, each curve relative to its own value at the label instant, with its maximum marked. The shaded band is the interval between the newest window’s occlusion-sensitivity centroid and the label instant. This interval corresponds to 0.61 px at the ninetieth percentile of label speed and 1.29 px at the ninety-ninth, against an annotation dispersion of 0.6353 px (Supplement Sec. 3). The curves deepen with speed, and all of them peak between the two lines.

ing centroid falls 25 ms before the instant t at which the label asserts a state. Both statements are read from the released source, and the 25 ms figure is a reference point rather than a property of the trained network.

3.3. The measured influence profile

The newest-window occlusion-sensitivity centroid is computed from the measured bin-ablation sensitivities of a released checkpoint. We take 480 samples of the released Gen1 validation split with the released rvt-t checkpoint, warm the recurrent state for eight steps, and occlude each bin in turn. For sample i , let Y_i denote the emitted detection tensor from the unoccluded input and $Y_i^{(-k)}$ the tensor obtained after zeroing temporal bin k while keeping the incoming recurrent state

identical. We define the influence of bin k as

$$s_k = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i^{(-k)} - Y_i\|_2}{\|Y_i\|_2}, \quad N = 480, \quad (1)$$

$$c_s = \frac{\sum_k s_k c_k}{\sum_k s_k},$$

where c_k is the center time of bin k relative to the label instant. The occluded forward pass receives the same incoming recurrent state as the corresponding reference pass, and its returned state is discarded. The profile is a property of the current window and not of a perturbed history.

The profile is balanced between the two halves of the window (Fig. 1a) and yields an occlusion-sensitivity centroid of -23.81 ms, compared with -25.00 ms for uniform weighting. The released configuration therefore places the measured centroid near the window reference without implying uniform network sensitivity. This is the current window’s occlusion-sensitivity profile with the recurrent state held fixed, not a decomposition of the full temporal support.

Recurrent support beyond the newest window. The detector is recurrent, so its temporal support does not end at the newest window. Occluding past windows and propagating their state to the current step shows that the newest accounts for 22.7% of the summed window-ablation sensitivity over a 1000 ms horizon. The tail does not decay to zero within that second, so the window-ablation-sensitivity centroid follows the horizon instead of converging: -161 ms over 500 ms of history and -299 ms over 1000 ms. The full recurrent temporal support is not identified by this sensitivity measurement.

Across the released checkpoints and a second architecture. RVT releases three Gen1 checkpoints differing only in embedding width, attention head dimension and FPN depth. SSM-ViT [25] releases two more in which each stage’s ConvLSTM is replaced by an S5 state space layer, at 9.68 and 18.19 M parameters against RVT’s 9.87 and 18.54. It distributes RVT’s own preprocessed Gen1 and inherits its window construction. On the same sequences, and driving each the way its own release drives it, all 5 checkpoints put the newest window’s occlusion-sensitivity centroid between -23.76 and -24.72 ms, none further than 1.24 ms, or 25% of one bin, from the uniform-weight value. The state the SSM release carries between evaluation chunks reaches none of its outputs. Under the released streaming evaluation, this bounds

the available recurrent history to at most the current 21-window chunk. It ranges from one to 21 windows with position in the chunk, as the supplement measures. The support behind that window is not fixed in the same way either: over 576 samples of the three RVT capacities, the newest window’s share of a 1000 ms horizon falls from 22.7% to 19.0% and the sensitivity centroid moves from -299 to -339 ms.

3.4. Regression-based temporal coefficient

The influence profile is a statement about the network’s input. Even a predictor with uniform local sensitivity can emit a state describing a later instant, and that quantity is measured from the same checkpoint’s own detections.

A detector reporting the state an object occupied τ earlier incurs the physical displacement $-\tau v$, where v is the latent true velocity. We observe only the finite-differenced label velocity $\tilde{v} = v + \eta$ and therefore fit Eq. 2 using $\tilde{v}$. Thus $\tau = -\tau_P$ on the axis of Sec. 3. With $\hat{c}$ the predicted center, c^* the label center, A the area of the ground-truth box, $\hat{u} = \tilde{v}/\|\tilde{v}\|$ and R a quarter turn, b is a two-dimensional intercept, fitted globally or as per-sequence intercepts in the reported specification,

$$\hat{c} - c^* = b - \tau \tilde{v} + \rho R \tilde{v} + \gamma \sqrt{A} \hat{u} + \varepsilon \quad (2)$$

is fitted by stacking both components of every match into one least squares problem, with standard errors clustered by sequence. It is the temporal calibration of [5] carried to a box center, taken as a slope rather than as the per-object ratio $-(\hat{c} - c^*, \hat{u})/\|\tilde{v}\|$, whose variance diverges as its denominator approaches zero. For comparison, the per-object ratio evaluated on the same rows has a mean of $+24.22 \pm 27.73$ ms, on the hypothesis evaluated in Sec. 3.5, with an interquartile range of 430 ms. Written this way the regression estimates a regression-based temporal coefficient under Eq. 2: no magnitude of the residual is taken, and τ does not load on a direction estimated from the same noisy velocity. The reported full specification adds four coordinate/size regressors and per-sequence intercepts to the terms displayed in Eq. 2. The coefficient is a temporal offset only if no velocity-proportional spatial bias remains after these controls. Such a bias is indistinguishable from τ in Eq. 2.

$R\tilde{v}$ is the label velocity rotated a quarter turn. Because a pure temporal offset acts along $\tilde{v}$ rather than $R\tilde{v}$, ρ serves as a negative-control coefficient. A nonzero value indicates residual spatial misspecification. $\sqrt{A} \hat{u}$ absorbs a displacement along travel proportional to box side, which is needed because that

Specification	τ (ms)	placebo ρ (ms)
velocity estimator, no geometric controls		
forward difference	-14.68 ± 4.99	-6.53 ± 1.68
backward difference	$+45.65 \pm 5.59$	-6.15 ± 1.71
centered difference	$+17.51 \pm 5.06$	-7.76 ± 1.92
controls added, centered difference throughout		
+ box side along travel	$+14.62 \pm 5.47$	-7.75 ± 1.91
+ four geometric terms	-7.62 ± 7.01	-4.12 ± 1.90
+ per-sequence bias	-2.40 ± 7.18	-2.66 ± 1.71

Table 1. The regression-based temporal coefficient τ of the released rvt-t checkpoint over 22 534 matched boxes of 376 Gen1 validation sequences, and the placebo ρ beside it, the coefficient on the rotated label velocity, which no temporal offset can reach. Standard errors are clustered by sequence. The forward and backward control fits are reported in Supplement Sec. 1.

residual grows with box size and box size correlates with label speed at $+0.139$ on this split.

The finite difference used for the velocity. $\tilde{v}$ is finite differenced from labels, and the choice of difference changes the answer. A forward difference $(c_{k+1}^* - c_k^*)/\Delta$ carries the label noise δ_k of the current center with weight $-1/\Delta$, and the residual $\hat{c} - c_k^*$ carries the same δ_k with weight -1 , so their covariance is $\text{Var}(\delta)/\Delta$ when τ is exactly zero. A backward difference carries δ_k with the opposite sign and produces the opposite artifact. The centered difference $(c_{k+1}^* - c_{k-1}^*)/(\Delta_f + \Delta_b)$ does not use c_k^* and removes the direct same-frame shared-noise term. Residual bias from temporally correlated annotation errors is not identified by this construction. Fitting all three is therefore a control on the estimator itself, calibrated before use. On 2000 synthetic tracks whose detector has no lag whatever, with label centers corrupted at the dispersion the supplement’s difference ladder measures on Gen1 (0.6515 px), the forward estimator returns -5.64 ms and the backward one $+5.84$ ms against a closed form of -5.63 ms, they sum to $+0.20$ ms, and removing the label noise returns all three to $+0.02$ ms. The same simulation recovers an injected lag at 0.92 of its size, the attenuation an errors-in-variables treatment predicts at 0.933 for this noise.

Regression dataset and matching. The released checkpoint is run over the Gen1 validation split with its recurrent state carried across steps, and its detections at each labeled frame are matched to the ground truth at an overlap of 0.3 or above. The estimate is conditional on detections matched at the nominal label time and applies only to matches satisfying this association criterion. Requiring both label neighbors, so that a centered velocity exists, and discarding the matches whose two neighbors coincide on the integer label lattice, leaves 22 534 of 35 523 over 376 sequences at a me-

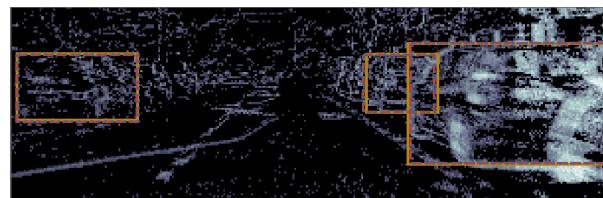
dian overlap of 0.873. Gen1’s track_id is a per-frame index rather than a persistent identifier, so association is rebuilt independently between adjacent label times. For each source box, candidate boxes of the same class are scored by IoU, the highest-scoring candidate is selected, and the link is retained only when IoU is at least 0.3. Ties follow the stored box order. This greedy rule does not enforce global one-to-one matching or carry identities across more than one interval. No speed floor is imposed. Accordingly, this coefficient characterizes the subset of detections that can be associated with ground truth at the nominal label time under the stated matching rule. It is not an unconditional timing estimate over all detector outputs.

Table 1 is a specification ladder, and the specification was chosen by the placebo rather than by τ . Under Eq. 2 with only the box-side term, $\tau = +14.62 \pm 5.47$ ms, but ρ is -7.75 ms at 4.0 standard errors, which indicates a term omitted from Eq. 2 that $R\tilde{v}$ is a proxy for. A split by the sign of the horizontal velocity locates it: ρ changes sign across that split, at -27.99 ms and $+21.84$ ms, which a rotational term would not do. Most label motion here is horizontal, so $R\tilde{v}$ maps a horizontal speed onto a vertical residual. A bias differing between oncoming and same-direction traffic is represented by this rotated regressor. Entering box height and image row along $\hat{y}$ and box width and image column along $\hat{x}$, and then a constant vector per sequence, returns the placebo to -2.66 ms at 1.6 standard errors. That is the specification reported, and the criterion that selected it is a coefficient on a regressor orthogonal to $\tilde{v}$.

Conditional on the reported specification, the regression-based temporal coefficient is $\tau = -2.40 \pm 7.18$ ms. Across the evaluated geometric-control specifications, τ ranges from -20.2 to $+11.1$ ms, whereas the occlusion-sensitivity centroid corresponds to $+23.81$ ms under the convention of Eq. 2. A cluster bootstrap over sequences puts the 95% interval at $[-17.0, +11.5]$ ms. Under the full control set, a perturbation injected directly along the fitted temporal regressor is recovered at 1.000 of its magnitude. The reported standard error and bootstrap interval condition on the selected specification and omit specification-selection uncertainty. Including all four geometric terms on both axes gives -3.62 ± 7.41 ms.

3.5. Separation between input sensitivity and regression-based temporal coefficient

The comparison combines measurements under different recurrent-state conditions. The centroid uses an eight-step warm-up with the incoming state fixed during ablation. The regression uses detections from the



released detections
ground truth at the label instant
ground truth at the newest window's centroid, $\delta = -23.81$ ms

Figure 2. One validation frame of the sweep of Fig. 1b: released detections (solid), ground truth at its label instant (dashed) and displaced to the newest window’s occlusion-sensitivity centroid (dotted). The frame is chosen by rule, the largest median label speed among frames with two labels above 20 px/s, its fastest label moves 45.2 px/s, so the two placements sit 1.08 px apart.

validation stream with the released state carried across frames. These are operational timing measurements of the same checkpoint, not estimates of a common full-recurrent-support centroid. Under the reported specification and its annotation-error assumptions, the point-estimate difference is 26.2 ms. The regression coefficient has a standard error of 7.2 ms. This is a regression-coefficient result rather than a latency estimate of this architecture. On the convention of Eq. 2 the centroid hypothesis is $\tau = +23.81$ ms, and no specification carrying geometric controls reaches it: τ spans -20.2 to $+11.1$ ms across them. Thus, 23.81 ms is the measured separation between the label instant and the newest-window occlusion-sensitivity centroid, whereas 26.2 ms is the point-estimate difference between that centroid and the reported regression coefficient. The centroid of the full recurrent support is not identified within the measured horizon. The standard error quoted is the output-time term’s. The variation across specifications is substantial relative to the reported standard error and is not captured by that standard error.

The detector is trained against the ground truth at the label time on a window ending there, so a measured offset near zero is consistent with the training objective indexed at the label time. The objective does not enforce the interval, and the instant the output describes and the centroid of the evidence used to produce it are different quantities that one benchmark timestamp stands for. At the ninetyth percentile of label speed on this split, 25.7 px/s, the 23.81 ms is 0.61 px of displacement, comparable to the annotation dispersion of the labels themselves (0.6353 px), and at the ninety-ninth it is 1.29 px. That occlusion-sensitivity centroid and the regression-based temporal coefficient are measurable from released artifacts alone.

Effect on benchmark mAP. The benchmark metric can be evaluated in its own units: we translate each ground-truth box center by $\delta\tilde{v}$, keeping its size and class fixed. This provides a first-order surrogate for temporal displacement under local constant velocity. The released predictions remain untouched. A detection carries no velocity of its own. Over 20 296 frames and 40 698 boxes, 68.7% of which carry a centered velocity and the rest are ignored. Ground-truth boxes without a centered velocity are marked ignored during evaluation, and detections matched to ignored boxes are excluded from TP/FP accounting. The resulting mAP against δ is smooth with its maximum at -10 ms (Fig. 1b). On the axis of Eq. 2 that is $\tau = +10$ ms, and mAP varies by at most 0.04 points over the interval. This descriptive curve does not by itself identify a preferred temporal value.

The full -60 to $+60$ ms sweep moves mAP by 0.9 points. On the -50 to $+30$ ms grid summarized in Table 2, the all-moving span is 0.37 points. Displacing the ground truth to the window’s centroid costs 0.0004 absolute mAP (0.04 percentage points, 0.12% relative to 0.3346). On the velocity-evaluable subset, standard average precision is nearly invariant over the measured label-to-centroid displacement. At the common -23.81 ms displacement, the ordering of all 5 released checkpoints is unchanged. Separately, scoring each checkpoint at its own preferred displacement leaves the reported ordering unchanged in the reported strata. A common displacement sweep reveals one localized inversion between rvt-b and S5-S in the 25–50 px/s stratum. Table 2 locates the flatness in that composition: 72.1% of the moving boxes travel below 10 px/s, where the displacement is 0.05 px, and the sweep deepens to 0.78 points between 25 and 50 px/s, where it is 0.77 px, past the labels’ own annotation dispersion of 0.6353 px (Supplement Sec. 3). The pooled response is composition-dependent: the mAP span is 0.37 points over all moving boxes but increases to 0.78 points in the 25–50 px/s stratum, where the centroid displacement costs 0.08 points.

4. A complementary within-exposure attribution limit

4.1. Exposure and labels

The predictor-side analysis concerns the temporal support of released detector inputs and outputs. DSEC provides a separate exposure-side analysis: published exposure intervals bound the interval within which event timing is observed, while illumination structure can introduce temporal variation unrelated to object motion. DSEC [12] publishes per-frame exposure meta-

speed (px/s)	share (%)	med. speed	centroid (px)	Δ mAP	span
all moving	100.0	4.0	0.10	0.04	0.37
< 10	72.1	2.1	0.05	0.02	0.11
10–25	19.2	15.0	0.36	0.01	0.33
25–50	7.5	32.3	0.77	0.08	0.78
≥ 50	1.2	57.4	1.37	0.00	0.37

Table 2. The sweep of Fig. 1b by object speed over the 27 943 ground-truth boxes with centered velocity. Other boxes are ignored. Δ mAP is the cost of shifting the ground truth to the newest-window occlusion-sensitivity centroid at -23.81 ms. Span is the mAP range over the -50 to $+30$ ms sweep. Both values are reported in points.

data, and the files resolve for eighteen train sequences. The width varies by a factor of 127 against a frame period fixed at 50 ms, and within one recording by up to a factor of 12.5. The auto-exposure ceiling of $14\,996\,\mu\text{s}$ holds 32.5% of frames, in six of the eighteen sequences, and the distribution is not continuous over that range (Fig. 3). This motivates treating ceiling exposure as a distinct stratum.

The released DSEC-Det labels comprise 390 118 boxes on 6693 tracks across 60 sequences, every label sits on the frame clock, and the median spacing between consecutive label times is $50\,001\,\mu\text{s}$. Over 361 628 triples of consecutive labels on a track, 7.1% of middle box centers fall within a hundredth of a pixel of their neighbors’ linear interpolant. This rules out exact local linear interpolation as a general description of the released trajectories without identifying the underlying annotation process. DSEC-Det ships more than one annotation version [15], the supplement names the one measured here.

Median box-center image speed is 40 px/s, with 360 px/s at the ninety-ninth percentile.

4.2. Mains-Frequency Signature and Per-Pixel Phase Test

Swiss mains is 50 Hz, and rectified lighting can produce 100.00 Hz intensity modulation. The observed event-rate spectrum contains the corresponding harmonic series because event cameras respond to log-intensity change. (Fig. 4). Sweeping 20–450 Hz on each whole recording, with no band fixed in advance and each frequency divided by the median of its own neighborhood, returns that series in every ceiling sequence: the ceiling median is 30 at 100.00 Hz, 125 at 200 Hz and 15 at 300 Hz, and the second harmonic is the strongest line in four of the six, never falling below 14.7. At the odd multiples of 50 Hz there is nothing: 1.8, 4.5 and 1.6 at 50, 150 and 250 Hz. This harmonic pattern is consistent with rectified 50-Hz illumination and is absent from the daytime control. The daytime sequence carries the fundamental at 0.5 and the second harmonic

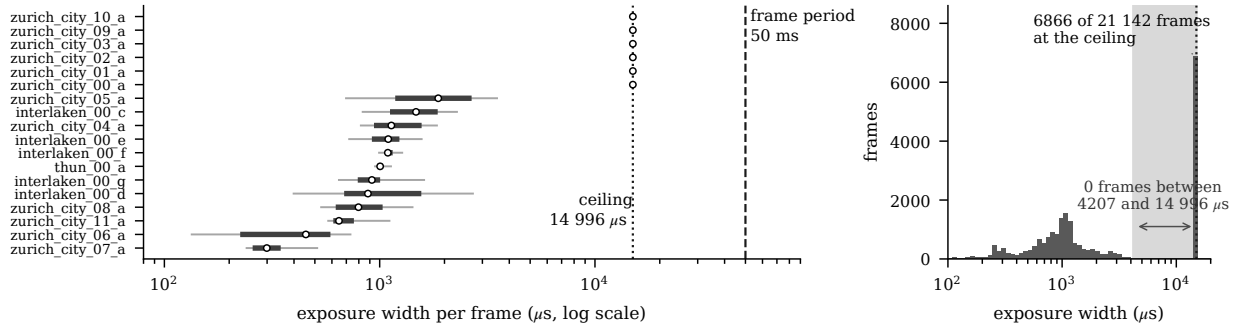


Figure 3. Published per-frame exposure width across the eighteen DSEC training sequences whose exposure files resolve: per sequence (left, log axis, median, interquartile range and 5–95% band, against the fixed 50 ms frame period) and pooled over 21 142 frames (right). 32.5% of frames sit at the auto-exposure ceiling of 14 996 μs and zero fall between the largest non-ceiling width of 4207 μs and that ceiling. This creates a separated ceiling-versus-non-ceiling exposure regime. The six ceiling sequences are the ones measured in Sec. 4.

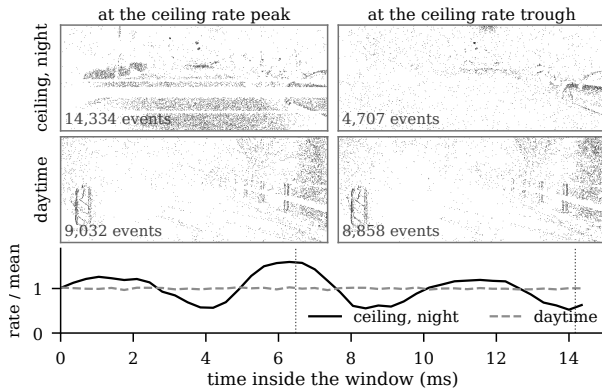


Figure 4. Top: 350 μs slices of the ceiling-exposure sequence `zurich_city_09_a` and of the daytime control at the instants when the ceiling sequence's own event rate reaches its peak and trough inside its densest published exposure window. Bottom: that rate in 350 μs bins, each sequence normalized to its own mean, the ceiling rate moves by a factor of 3.0 between its extremes, the daytime rate by 1.1.

at 6.2, against 14.7 for the weakest ceiling sequence.

Per pixel, the Rayleigh statistic $Z = R^2/N$ is $\text{Exp}(1)$ under independent uniform phases. The off-frequency and daytime controls empirically calibrate this nominal null, while the ceiling sequence has median $Z = 2.935$ at 100.00 Hz and 24.07% of live pixels reach $p < 10^{-3}$.

4.3. All six ceiling sequences, per labeled box

The same statistic taken per labeled box, over the events inside that box during that frame's own exposure window, extends to every ceiling sequence, the supplement tabulates all six of them over 16 516 boxes. Across the six ceiling sequences, 95.4–100.0% of boxes satisfy the locked criterion defined in Supplement Ta-

ble 2.

4.4. Within-box temporal dispersion under that signature

That signature limits attribution of within-exposure event timing to object timing. Within-box dispersion is 206.6 μs against a null of 93.0 μs , with an excess of 183.6 μs . A frame-level permutation preserving pooled temporal distributions and box counts gives a null of 78.6 μs . The excess remains after the clean-pixel control, while scene and illumination components remain unresolved.

5. Limitations

Results use the released artifacts, local occlusion sensitivity, nominal-time greedy matching, the reported specification, and local constant velocity. Correlated annotation-error bias and the full recurrent centroid remain unidentified. Gen1 and DSEC are separate case studies. DSEC dispersion is not object-specific.

6. Conclusion

A nominal benchmark timestamp is not a temporal specification. The predictor exhibits distinct input-side sensitivity and a regression-based temporal coefficient on nominal-time-matched detections, while the DSEC case study shows that illumination structure limits the attribution of within-exposure event timing to object motion.

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